

# Portraits of Automated Facial Recognition : Deep Learning Methods for Face detection and recognition

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Face recognition technology has become increasingly popular in recent years. Initially defined as a branch of *artificial intelligence* that allow robots to perceive visually, this new technology is now part of a more generalized discipline of *computer vision*. Many applications of image recognition have been introduced in different areas of daily life. We can apply image recognition for example not only to standard pictures but to analyse images spanning a more broad range of the *electromagnetic spectrum*. We know for example that X-rays are used in medicine to take photographs inside the human body without the need of surgical intervention. The *gamma rays* and *radio waves* in MRI can capture images of thin sections of the human body used to diagnose and treat a variety of diseases [1]. In this article, we will focus on analyzing different methods that are implemented to draw automated portraits of face recognition.

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## 1. INTRODUCTION

Facial recognition technology offers a host of benefits compared to other biometric methods such as fingerprint and iris: in addition to being natural and non-intrusive, the most important advantage of using the face as a metric is that it can be captured remotely and with discretion. Considering the six biometric attributes analysed by [2], facial features scored the highest in custom machine-readable passport control system MRTD<sup>1</sup> based on a number of assessment factors such as registration, renewal and audience requirements. For that matter, Facial Recognition among biometric technologies, has become increasingly important due to rapid advances in imaging devices (such as surveillance cameras, mobile phones etc ...) and the availability of huge amounts of images on the web in addition to increased security requirements.

Face detection considers locating the face in an image without identification. Although many algorithms of face detection existed before 2001, a major breakthrough in face

detection algorithms came with Viola-Jones paper "Rapid Object Detection using a Boosted Cascade of Simple Features" [3, 4]. Unlike previous facial detection methods that relied on pixel analysis, Viola-Jones devised an algorithm called "Haar-classifier" that relied on Haar-like features.

Haar's classifier is a machine learning algorithm trained with many positive and negative samples to detect objects in images. For the classifier to work properly, the size of the image in the training set must be the same as the size of the input image used for object detection.

## 2. ALGORITHMES

### A. Appearance-based algorithms

Appearance-based algorithms use pixel data in images as a main method for recognition. On the other hand, direct correlation, Eigen-face and Fisher-Face belong also to this class of algorithms. Direct correlation uses the direct comparison of image pixels of two facial images, producing a similarity scores [5, 6]. Unlike a direct correlation method which uses facial images in their original image space, the Eigen-face and Fisher-face algorithms reduce the image to the most discriminating factor and perform the comparison between the images in reduced size space [7, 8].

### B. Algorithms based on active appearance

Active appearance-based algorithms contain statistical information about the shape of an image and texture variation. Fouques et al. [9] applied principal component analysis (PCA) to generate a statistical model that locates reference points on a set of training images. Reference points are used to train the model to learn the translation between the synthesized model parameter and the test images. To find the image pairs and match them to each other, the residual error of the model is measured against the predicted changes in the current parameters of the model which leads to a better fit and a better recognition [10].

### C. Support Vector Machines

Support vector machines use a set of training images to calculate the optimal separation hyperplane. Guo et al. [11] applied this method to facial recognition using a binary tree classification, where facial images are iteratively classified as belonging to one

<sup>1</sup>[https://www.icao.int/meetings/tag-mrtd/tagmrtd22/tag-mrtd-22\\_wp24.pdf](https://www.icao.int/meetings/tag-mrtd/tagmrtd22/tag-mrtd-22_wp24.pdf)

of two classes that propagate in a binary tree structure until that a final classification decision can be made.

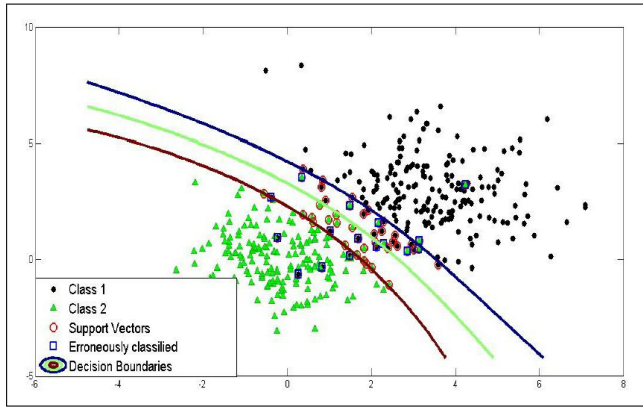


Fig. 1. SVM Classifier.

#### D. Bayesian algorithms

The Bayesian model algorithms show a probabilistic measure of similarity derived from a Bayesian analysis of the difference between facial images. Computing probability functions using image intensity differences between datasets leads to a similarity score from Bayes' rule, which can then be used for facial recognition classification [12].

#### E. Neural Networks

Neural networks provide information processing, which is similar to how information is processed in biological systems such as the human brain. Their main strength is the ability to learn from examples, fault tolerance and robustness. They are suitable for recognizing faces that have a lot of variation and yet require little modification of the recognition algorithm. Lawrence [13] describes how to train a neural network classifier for facial identification and recognition.

#### F. Texture Based Algorithms

Texture-based algorithms extract texture in images, dividing the face into multiple regions. Local Binary Model (LBP) is an example of texture-based algorithms, where weighted LBP features are extracted to generate a vector of different features. Two LBP feature vectors are matched by applying a weighted Chi-square distance measure [14].

#### G. Feature-Based Algorithms

These algorithms extract a set of geometric features such as distance from facial images and use these features as a reference to compare between the images. Local feature analysis is one example of feature-based algorithms [15].

### 3. FACIAL RECOGNITION WITH KERAS

In this chapter we use a convolutional neural network (CNN) to extract the characteristics of input images. The network follows the approach described in [16] with modifications inspired by the OpenFace project. Keras is used to implement CNN, Dlib

and OpenCV to align faces to input images. Facial recognition performance is evaluated on a small subset of the LFW dataset, which can be replaced with a custom dataset, for example, images of students enrolled in the university (for biometric verification).

#### A. Example implementation of the algorithm using SVM

In the example of Figure 3, we use the SVM classifier to test face recognition algorithm.

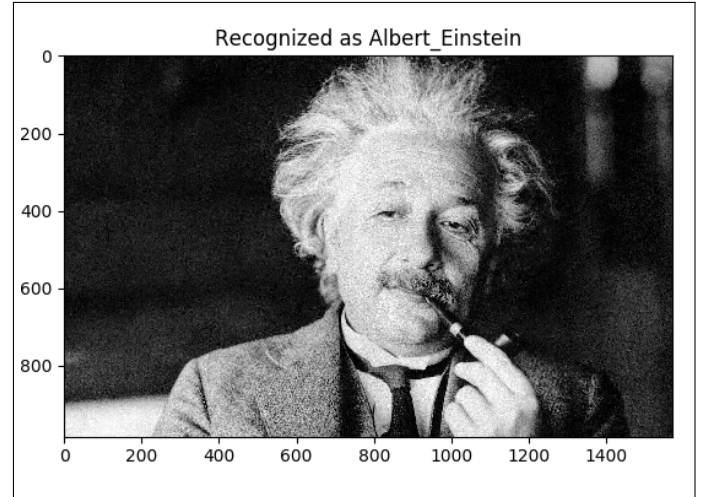


Fig. 2. Image of Albert Einstein identified using the SVM algorithm.

### 4. CNN ARCHITECTURE AND TRAINING

The CNN architecture used here is a variant of the inception architecture [17]. More precisely, it is a variant of the CNN architecture described in [16] and identified as model nn4.small2 which is named here model.cnn in the OpenFace project. In this chapter, we consider Keras implementation of this model described in the Keras-OpenFace project. The details of the architecture are not detailed here, it is only useful to know that there is a fully connected layer with 128 hidden units followed by an L2 normalization layer above the convolutional base. These top two layers are called the wrap layer from which the 128-dimensional wrap vectors can be obtained. The complete model can be found in model.py.

Model training aims to learn an embedding  $f(x)$  of an image  $x$  such that the  $L_2$  squared distance between all faces of the same identity is small and the distance between a pair of faces of different identities is large. This can be achieved with a triplet loss  $L$  which is minimized when the distance between an anchor image  $x_i^a$  and a positive image  $x_i^p$  (of the same identity) in the integration space is less than the distance between this anchor image and a negative image  $x_i^n$  (different identity) of at least one margin  $\alpha$ .

$$L = \sum_{i=1}^m [\|f(x)_i^a - f(x)_i^p\|_2^2 - \|f(x)_i^a - f(x)_i^n\|_2^2 + \alpha]_+ \quad (1)$$

During the training phase, it is important to select the triplets whose positive  $(x_i^a, x_i^p)$  and negative  $(x_i^a, x_i^n)$  pairs are

difficult to discriminate. That is, their distance difference in the embedding space must be less than the margin  $\alpha$  otherwise the network is unable to learn useful integration. Therefore, each training iteration should select a new batch of triples based on the integrations learned in the previous iteration.

### A. Distance threshold

To find the optimal value of  $\tau$ , face verification must be evaluated over a range of distances. At a given threshold value, all the pairs of possible integration vectors are classified either having the same identity or having a different identity and the whole compared to the expected result. Since we are dealing with asymmetric classes (much more negative pairs than positive pairs), we use the F1 score as the evaluation metric instead of accuracy.

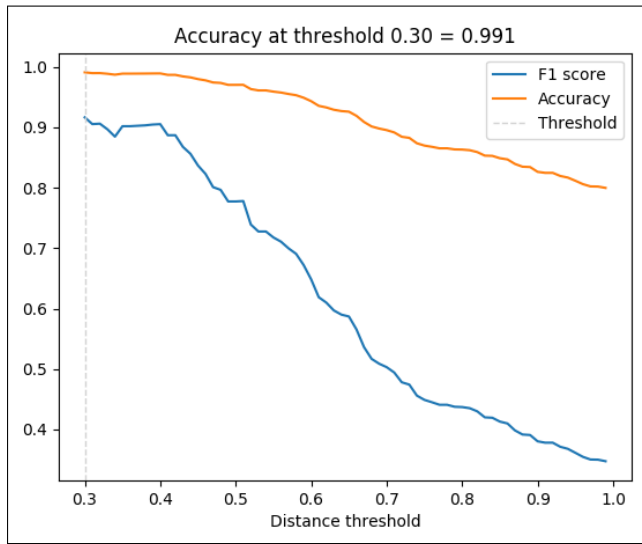


Fig. 3. Distance Threshold.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{TP}{TP + \frac{1}{2} \times (FP + FN)} \quad (2)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

The facial recognition algorithm implemented here has an accuracy of 0.991 at  $\tau = 0.30$ .

That's not bad given a baseline of 0.89 for a classifier that always predicts a different identity (there are 450 positive pairs and 3000 negative pairs), but since nn4.small2.v1 is a relatively small model, it is always lower than what can be achieved by the best models (>0.99).

### B. KNN Classification

Given an estimate of the distance threshold  $\tau$ , face recognition is now as simple as calculating the distances between a given input integration vector and all integration vectors in a database. The input image is assigned a label (i.e. an identity) of input image from the database with the smallest distance if it is less than  $\tau$  or an unknown label otherwise.

A more robust approach is to label the input data using the highest-scoring entries in the database, which essentially represents a KNN classification with Euclidean distance metric. Alternatively, a linear support vector machine (SVM) can be trained with the database entries and used to classify and identify new entries. During the training phase, we use 50 percent of the dataset and 50 percent for the evaluation.

### C. KNN Algorithm

#### Algorithm 1. KNN algorithm

```

1: procedure KNN( $A, B, a, b$ ) ▷ train and test data
2:    $knn \leftarrow KNeighborsClassifier(nneighbors = 1, metric = 'euclidean')$ 
3:    $svc \leftarrow LinearSVC()$ 
4:    $mod1 \leftarrow knn.fit(A, a)$ 
5:    $mod2 \leftarrow svc.fit(A, a)$ 
6:    $KNN \leftarrow accuracy.score(b, knn.predict(B))$ 
7:    $SVC \leftarrow accuracy.score(b, svc.predict(B))$ 
8:   return KNN, SVC ▷ Accuracy Scores

```

$A : X.train, B : X.test, a : y.train, b : y.test$

### D. Results

In our example we see that the KNN classifier achieves an accuracy of 0.96 on the test set while the SVM classifier achieves an accuracy of 0.98.

Although both algorithms produce satisfactory results in terms of score and accuracy. The SVM algorithm offers a better result with a much higher classification speed than KNN.

As a matter of fact, the KNN algorithm uses a voting system to determine to which class an unclassified object belongs, taking into account the class of its nearest neighbors. The SVM algorithm on the other hand is extremely fast, able of classifying 12 megapixel aerial images in about ten seconds (unlike KNN which takes between forty and fifty seconds).

Overall, the KNN algorithm classifies accurately; however, it generates several small classification errors that interfere with the final classified image. On the other hand, the SVM algorithm will sometimes misclassify a large object which rarely interferes with the final classified image.

## 5. STUDY CONCLUSION

Facial recognition is a biometric recognition method with practical features in national and social security fields.

With the deepening of computer vision research, small-scale face recognition has achieved good performance results. On the other hand, with the advent of Big Data technologies, small-scale facial recognition methods are no longer able to meet social needs. As a result, the analysis of the performance and accuracy of facial recognition algorithms applied to big data has become the new focus of research in the domain of artificial intelligence.

## 6. OUVERTURE ET PERSPECTIVES

The recent years have witnessed a growing interest in computer vision as well as machine learning and in particular in the fields

of security, multimedia and digitalisation. Research in the field of facial recognition and image processing began in the 1960s, but only recently, a significant upsurge in the development of computer vision systems has become prominent.

Although precise and accurate facial recognition system still offers many challenges for researchers at this time, the rise of facial recognition technology comes in fact with a wider range of functions.

Indeed, not only computer vision technologies are easier to implement nowadays, but they will also be able to enhance images in challenging weather and light conditions. These technologies can be implemented alongside artificial intelligence to create a more powerful applications. For example, computer vision can be implemented in vital applications such as the development of AGI artificial general intelligence and ASI artificial super-intelligence by improving and processing perceived visual information in a manner that is inspired by the human perceptive and neuro-visual system.

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